

Statistics Review 2

Confidence Intervals and Hypothesis Testing

Tu Li

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Econ 453 University of Arizona

Roadmap

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2. The Confidence Interval
3. The Large-Sample Approach
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1. From Point to Set Estimation

1.1 Point vs Set Estimation

Lecture 3 gave us a **point estimate** of the population mean:

$$\hat{\mu} = \bar{X}.$$

One number. A single best guess.

But a single number tells you *nothing about how confident you should be*. If a survey of $n = 100$ workers gives $\bar{X} = \$52,000$:

- Is the population mean probably between \$51K and \$53K?
- Or could it easily be as low as \$45K or as high as \$60K?

We can't tell from the point estimate alone.

Solution: report a **range of plausible values**. This is a **set estimate** (or **interval estimate**). The canonical example is the **confidence interval** — the main subject of this lecture.

Once we have CIs in hand, a closely related procedure — **hypothesis testing** — lets us *test specific claims* about μ (“Is the mean equal to 50K?”). That’s the second half of the lecture.

2. The Confidence Interval

2.1 The Concept

Definition

A $(1 - \alpha) \cdot 100\%$ **confidence interval** for μ is a *random* interval $[\hat{L}, \hat{U}]$, computed from the sample, with the property

$$\mathbb{P}(\hat{L} \leq \mu \leq \hat{U}) = 1 - \alpha.$$

Notation reminder. α (Greek “alpha”) is the **miss rate**: the probability that the interval fails to cover μ . Common choices:

- $\alpha = 0.05 \Rightarrow 95\%$ CI (the default).
- $\alpha = 0.01 \Rightarrow 99\%$ CI (more conservative).
- $\alpha = 0.10 \Rightarrow 90\%$ CI (less conservative).

Key point. The interval is random — both endpoints depend on the sample. The population mean μ is *not* random; it’s a fixed unknown. So the probability $1 - \alpha$ refers to the procedure: across many samples, how often does the interval succeed in catching μ ?

2.2 Interpretation: The Repeated-Sampling Story

Imagine drawing many independent samples, computing a 95% CI from each. The intervals will differ from sample to sample — they're random. But the procedure has a guarantee:

About 95% of these intervals will contain the true μ .

The other 5% will miss.

The “95%” is a property of the *procedure*, not of any particular interval. Once you compute a specific CI like [48,472, 55,528], the true μ either is or isn't in it — no randomness left.

Practical wording. When you report a 95% CI, you say:

“We are 95% confident that μ lies in [48,472, 55,528].”

“Confident” (a hedged word about the method) replaces “probability” (which would be wrong).

2.3 Common Misconception

Wrong

“There is a 95% chance that μ lies in $[48,472, 55,528]$.”

Why is this wrong? Once the interval is computed, both endpoints are fixed numbers and μ is a fixed number. Either μ is in the interval or it isn't — there's no probability involved.

Right

“If we repeated the sampling procedure many times, about 95% of the resulting CIs would contain μ .”

The probability statement applies to the *procedure*, not to any *single* interval already computed.

(You will sometimes hear the loose wording in textbooks and papers — everyone does it. But it's worth understanding what's really going on.)

3. The Large-Sample Approach

3.1 Why We Skip the t-Distribution

Traditional statistics courses handle the case “ σ is unknown” by introducing the **Student's t distribution**: a bell curve slightly fatter in the tails than $N(0, 1)$, indexed by degrees of freedom.

In this course we take a different route. Modern data-science applications usually have **large samples** — hundreds or thousands of observations — and in that regime, the distinction between the t and the Normal becomes negligible.

So we will use the **Normal distribution throughout**, with a rule of thumb on n . This keeps all our inference machinery on a single distribution.

3.2 The Rule of Thumb

Large-sample approximation

For $n \geq 60$, the standardized sample mean

$$\frac{\bar{X} - \mu}{s/\sqrt{n}} \underset{\text{approx}}{\sim} N(0, 1)$$

to a very good approximation, *even when σ is unknown and we plug in the sample standard deviation s .*

Two ingredients.

- **CLT** (Lecture 3 §6): for moderate n , the distribution of $\bar{X}$ is approximately Normal regardless of the population shape.
- **Plug-in works**: for $n \geq 60$, the sample SD s is a tight enough estimate of σ that replacing σ by s barely shifts the distribution.

Together: the standardized statistic with s in place of σ is well approximated by $N(0, 1)$ when n is at least 60.

3.3 The Convention We'll Use

From now on:

- If σ is **known**, use $\sigma/\sqrt{n}$ for the standard error.
- If σ is **unknown**, use $s/\sqrt{n}$ instead.

In both cases, all critical values come from the **standard Normal table** (or software). No t-tables, no degrees of freedom.

Where this shows up later. OLS coefficient inference in Lecture 7 takes the same large-sample view: report a coefficient estimate $\hat{\beta}$ and its estimated standard error, and treat $\hat{\beta}/\widehat{SE}(\hat{\beta})$ as approximately Normal. No t-stat distinct from a Z.

4. Constructing a Confidence Interval

4.1 The General Formula

Key Result — CI for μ

$$\text{CI} = \bar{X} \pm z_{\alpha/2} \cdot \frac{s}{\sqrt{n}}$$

(Use $\sigma/\sqrt{n}$ instead of $s/\sqrt{n}$ if σ is known.)

Notation reminder. $z_{\alpha/2}$ is the **critical value**: the number such that

$$\mathbb{P}(Z > z_{\alpha/2}) = \alpha/2 \quad \text{where } Z \sim N(0,1),$$

i.e., $\alpha/2$ of the Normal mass lies to its right. Equivalently

$$\Phi(z_{\alpha/2}) = 1 - \alpha/2.$$

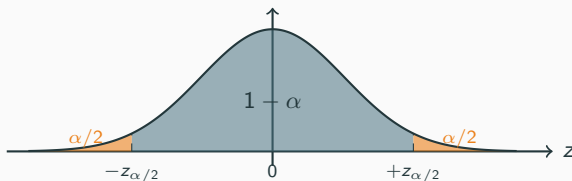
The quantity $z_{\alpha/2} \cdot s/\sqrt{n}$ is called the **margin of error**, often written ME. The CI is “estimate $\pm$ margin of error.”

4.2 Z Critical Values

Three values to memorize:

Confidence	α	$z_{\alpha/2}$
90%	0.10	1.645
95%	0.05	1.96
99%	0.01	2.576

The picture for the 95% case:



Central $1 - \alpha$ region: where Z falls with high probability.

4.3 Why This Formula Works

We claim the random interval $\bar{X} \pm z_{\alpha/2} \cdot \sigma / \sqrt{n}$ has probability exactly $1 - \alpha$ of containing the true μ . Let's see why, step by step.

Step 1: start with the standardized sample mean. By the CLT (Lecture 3 §6, or the exact-Normal case in Lecture 3 §4.2), for an i.i.d. sample with known σ ,

$$Z = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}} \sim N(0, 1).$$

Step 2: use the definition of $z_{\alpha/2}$. By construction, the central $1 - \alpha$ region of the standard Normal lies between $\pm z_{\alpha/2}$:

$$\mathbb{P}(-z_{\alpha/2} \leq Z \leq z_{\alpha/2}) = 1 - \alpha.$$

4.3 Why This Formula Works 2

Step 3: substitute the definition of Z .

$$\mathbb{P}\left(-z_{\alpha/2} \leq \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq z_{\alpha/2}\right) = 1 - \alpha.$$

Step 4: multiply all three parts by $\sigma/\sqrt{n}$. This is a positive number, so inequalities don't flip:

$$\mathbb{P}\left(-z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} \leq \bar{X} - \mu \leq z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha.$$

4.3 Why This Formula Works 3

Step 5: subtract $\bar{X}$ from all three parts.

$$\mathbb{P}\left(-\bar{X} - z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} \leq -\mu \leq -\bar{X} + z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha.$$

Step 6: multiply by -1 . This *does* flip the inequalities (and the order):

$$\mathbb{P}\left(\bar{X} - z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha.$$

4.3 Why This Formula Works 4

Conclusion

The random interval

$$\left[\bar{X} - z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}, \bar{X} + z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} \right]$$

contains μ with probability $1 - \alpha$. That's the formula on the previous slide. ✓

When σ is unknown. The same chain of steps works with s in place of σ — except Step 1 becomes *approximate* rather than exact. By the large-sample rule (§3.2),

$$\frac{\bar{X} - \mu}{s/\sqrt{n}} \underset{\sim}{\text{approx}} N(0, 1) \quad \text{for } n \geq 60.$$

4.3 Why This Formula Works 5

Carrying the approximation through Steps 2–6, the resulting CI contains μ with probability *approximately* $1 - \alpha$. In the large-sample regime, the approximation is excellent.

4.4 Step-by-Step Recipe

How to build a CI for μ

1. Compute $\bar{X}$ (and s , if σ is unknown).
2. Pick confidence level $1 - \alpha$ and look up $z_{\alpha/2}$.
3. Compute the standard error: $SE = \sigma/\sqrt{n}$ if σ is known, or $\hat{SE} = s/\sqrt{n}$ if it is unknown.
4. Compute the margin of error: $ME = z_{\alpha/2} \cdot SE$.
5. Report: $CI = [\bar{X} - ME, \bar{X} + ME]$, with an interpretation in plain English.

The recipe is the same regardless of whether σ is known. The *only* difference is which standard-error formula to plug in.

Two examples to follow.

5. CI Worked Examples

5.1 Example 1: Light-Bulb Lifetimes (σ known)

Setup. A manufacturer tests $n = 64$ light bulbs and measures their lifetimes. From historical data the population standard deviation is known: $\sigma = 100$ hours. The sample mean is $\bar{X} = 350$ hours. Build a 95% CI for μ .

Step 1: read off $\bar{X}$. $\bar{X} = 350$.

Step 2: critical value. For 95% confidence, $\alpha = 0.05$ and $z_{\alpha/2} = 1.96$.

Step 3: standard error. σ is known, so

$$SE = \frac{\sigma}{\sqrt{n}} = \frac{100}{\sqrt{64}} = 12.5.$$

5.1 Example 1: Light-Bulb Lifetimes (σ known) 2

Step 4: margin of error.

$$ME = z_{\alpha/2} \cdot SE = 1.96 \times 12.5 = 24.5.$$

Step 5: report and interpret.

$$CI = [350 - 24.5, 350 + 24.5] = [325.5, 374.5] \text{ hours.}$$

Interpretation. We are 95% confident that the population mean lifetime of these bulbs is between 325.5 and 374.5 hours.

5.2 Example 2: Annual Income (σ unknown)

Setup. A city wants to estimate average annual income of its workers. A survey of $n = 100$ randomly chosen workers yields $\bar{X} = \$52,000$ and $s = \$18,000$. Build a 95% CI for μ .

Step 1: read off $\bar{X}$ and s . $\bar{X} = 52,000$; $s = 18,000$.

Step 2: critical value. $z_{\alpha/2} = 1.96$.

Step 3: standard error. σ is unknown, but $n = 100 \geq 60$ — large-sample regime, so plug in s :

$$\hat{SE} = \frac{s}{\sqrt{n}} = \frac{18,000}{\sqrt{100}} = 1,800.$$

5.2 Example 2: Annual Income (σ unknown) 2

Step 4: margin of error.

$$ME = 1.96 \times 1,800 = 3,528.$$

Step 5: report and interpret.

$$CI = [52,000 - 3,528, 52,000 + 3,528] = [\$48,472, \$55,528].$$

Interpretation. We are 95% confident that the average worker in this city earns between \$48,472 and \$55,528 per year.

6. Hypothesis Testing — Concepts

6.1 The Setup

A **hypothesis test** is a decision procedure: use a sample to pick between two competing claims about the population.

Definition

- **Null hypothesis** H_0 : the “default” claim, usually a specific value like $\mu = \mu_0$.
- **Alternative hypothesis** H_1 : the claim we’d accept if the data contradicts H_0 .

The procedure is **asymmetric**: we *reject* H_0 only when the data is unusual under H_0 . Otherwise we *fail to reject*.

Notation reminder. “Fail to reject” is not the same as “accept” — it means the evidence wasn’t strong enough to overturn the default. The defendant in a trial is found “not guilty,” not “innocent.”

6.2 One-Sided vs Two-Sided

Three standard forms for H_1 when testing $H_0 : \mu = \mu_0$:

- **Two-sided:** $H_1 : \mu \neq \mu_0$ — “the mean is different (in either direction).”
- **Right-tailed:** $H_1 : \mu > \mu_0$ — “the mean is larger.”
- **Left-tailed:** $H_1 : \mu < \mu_0$ — “the mean is smaller.”

Which one to use? Depends on the substantive question:

- “Is mean reaction time different from 2 seconds?” → two-sided (could differ in either direction).
- “Is mean reaction time below 2 seconds?” → left-tailed.
- “Is mean click-through above its benchmark?” → right-tailed.

Two-sided is the default when in doubt. It's the more conservative choice.

6.3 Type I and Type II Errors

A hypothesis test can go wrong in two ways:

	H_0 true	H_0 false
Reject H_0	Type I error	✓ correct
Fail to reject	✓ correct	Type II error

- **Type I error**: rejecting a true null. “False alarm.”
- **Type II error**: failing to reject a false null. “Missed detection.”

Significance level

The **significance level** α is the Type-I error rate we’re willing to tolerate:

$$\alpha = \mathbb{P}(\text{Type I error}) = \mathbb{P}(\text{reject } H_0 \mid H_0 \text{ true}).$$

Standard choices: $\alpha = 0.05$ (default), $\alpha = 0.01$ (stricter). Smaller α = harder to reject = fewer false alarms, but more missed detections.

7. Hypothesis Testing — Construction

7.1 The Test Statistic

Under $H_0 : \mu = \mu_0$, the **test statistic**

$$Z = \frac{\bar{X} - \mu_0}{s/\sqrt{n}}$$

is approximately $N(0, 1)$ for $n \geq 60$ (Lecture 3 CLT + large-sample plug-in).

Interpretation. Z counts how many standard errors away from μ_0 the observed $\bar{X}$ lies.

- Small $|Z|$ (≤ 1 or so): $\bar{X}$ is close to μ_0 . No reason to doubt H_0 .
- Large $|Z|$ (≥ 2 or so): $\bar{X}$ is far from μ_0 . H_0 looks unlikely.

“How large is large enough to reject?” — the next two slides give two equivalent answers.

7.2 Critical-Value Method

Step-by-step recipe (critical-value method)

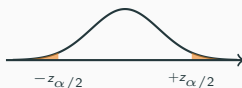
1. State H_0 and H_1 .
2. Pick α (default 0.05).
3. Compute test statistic $Z = (\bar{X} - \mu_0)/(s/\sqrt{n})$.
4. Find the critical value(s):
 - Two-sided: $\pm z_{\alpha/2}$ (e.g., ± 1.96).
 - Right-tailed: $+z_{\alpha}$ (e.g., $+1.645$).
 - Left-tailed: $-z_{\alpha}$ (e.g., -1.645).
5. Decision: reject H_0 if Z falls in the rejection region (beyond the critical value); otherwise fail to reject. State the conclusion.

7.2 Critical-Value Method 2

Rejection regions visualized:



Left-tailed
 $H_1 : \mu < \mu_0$



Two-tailed
 $H_1 : \mu \neq \mu_0$



Right-tailed
 $H_1 : \mu > \mu_0$

The shaded region in each panel is where H_0 gets rejected.

7.3 P-Value Method

An equivalent but more flexible way to make the same decision.

Definition

The **p-value** is the probability, *under* H_0 , of observing a test statistic at least as extreme as the one we got:

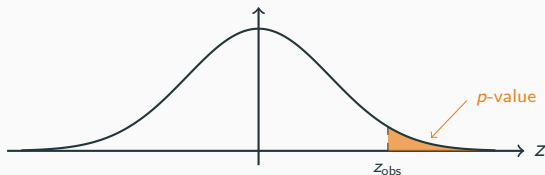
- Two-sided: $p = 2 \cdot \mathbb{P}(Z > |z_{\text{obs}}|)$.
- Right-tailed: $p = \mathbb{P}(Z > z_{\text{obs}})$.
- Left-tailed: $p = \mathbb{P}(Z < z_{\text{obs}})$.

Decision rule: reject H_0 if $p \leq \alpha$.

How to read it. Small p-value $\Rightarrow$ the observed data is very unlikely under $H_0 \Rightarrow$ reject H_0 . Large p-value $\Rightarrow$ the data is consistent with $H_0 \Rightarrow$ fail to reject.

7.3 P-Value Method 2

P-value picture (right-tailed case):



The shaded tail area is the p-value. (Two-sided: shade both tails beyond $\pm|z_{\text{obs}}|$ and sum.)

Why both methods? The critical-value method is geometric and easy to draw. The p-value method gives a continuous measure of evidence — a p-value of 0.001 is much stronger evidence against H_0 than $p = 0.04$, even though both reject at $\alpha = 0.05$. Software output (e.g., regression tables in Lecture 7) reports p-values.

8. HT Worked Examples

8.1 Example 1: Cereal Weight (σ known)

Setup. A cereal box should weigh $\mu_0 = 368$ g (label weight). A random sample of $n = 25$ boxes yields $\bar{X} = 372.5$ g. From the factory specs, $\sigma = 15$ g is known. Test, at $\alpha = 0.05$, whether the mean box weight differs from 368 g.

Caveat. $n = 25$ is below the $n \geq 60$ rule, but here σ is *known* — so we are *not* relying on the $s \rightarrow \sigma$ approximation. Provided the population is approximately Normal (a reasonable assumption for factory weights), the Z statistic is valid for any n .

Step 1: hypotheses.

$$H_0 : \mu = 368 \quad H_1 : \mu \neq 368 \quad (\text{two-sided}).$$

Step 2: $\alpha = 0.05$.

8.1 Example 1: Cereal Weight (σ known) 2

Step 3: test statistic.

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} = \frac{372.5 - 368}{15/\sqrt{25}} = \frac{4.5}{3} = 1.5.$$

Step 4 (critical-value method). Critical values: $\pm z_{0.025} = \pm 1.96$. The observed $Z = 1.5$ is in $[-1.96, 1.96]$, so we **fail to reject** H_0 .

Step 5 (p-value method, equivalent).

$$p = 2 \cdot \mathbb{P}(Z > 1.5) = 2 \cdot (1 - 0.9332) = 2 \cdot 0.0668 = 0.134.$$

Since $p = 0.134 > 0.05$, **fail to reject** H_0 .

Conclusion. The sample does *not* provide enough evidence to claim that the mean weight differs from 368 g.

8.2 Example 2: Drug Trial (σ unknown)

Setup. A clinical trial uses $\mu_0 = 2.0$ seconds as the benchmark mean reaction time. A sample of $n = 100$ patients gives $\bar{X} = 1.4$ seconds and $s = 2.0$ seconds. Test, at $\alpha = 0.01$, whether the trial population's mean differs from the benchmark.

Step 1: hypotheses.

$$H_0 : \mu = 2.0 \quad H_1 : \mu \neq 2.0 \quad (\text{two-sided}).$$

Step 2: $\alpha = 0.01$.

Step 3: test statistic. $n = 100 \geq 60$, so use Normal with s for σ :

$$Z = \frac{1.4 - 2.0}{2.0/\sqrt{100}} = \frac{-0.6}{0.2} = -3.0.$$

8.2 Example 2: Drug Trial (σ unknown) 2

Step 4 (critical-value method). Critical values: $\pm z_{0.005} = \pm 2.576$.

The observed $Z = -3.0$ is *beyond* -2.576 (in the rejection region), so we **reject** H_0 .

Step 5 (p-value method, equivalent).

$$p = 2 \cdot \mathbb{P}(Z > 3.0) = 2 \cdot 0.00135 = 0.0027.$$

Since $p = 0.0027 < 0.01$, **reject** H_0 .

Conclusion. Strong evidence that the trial population's mean reaction time differs from the 2.0-second benchmark. The sample mean is about 0.6 seconds below the benchmark.

Where this shows up later. Lecture 7 will use exactly this procedure to test whether a regression coefficient $\hat{\beta}$ differs from zero — the most-used hypothesis test in all of statistics and econometrics.

9. The CI $\leftrightarrow$ HT Equivalence

9.1 Two Sides of the Same Coin

Equivalence

For a two-sided test of $H_0 : \mu = \mu_0$ at level α :

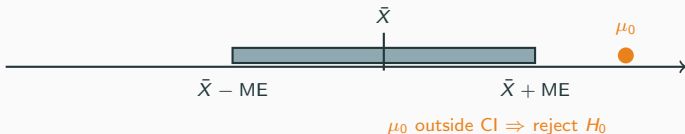
reject $H_0 \iff \mu_0$ lies **outside** the $(1 - \alpha) \cdot 100\%$ CI for μ .

Why? Both procedures compare $|\bar{X} - \mu_0|$ to $z_{\alpha/2} \cdot \text{SE}$:

- HT rejects iff $|\bar{X} - \mu_0| / \text{SE} > z_{\alpha/2}$, i.e., $|\bar{X} - \mu_0| > z_{\alpha/2} \cdot \text{SE}$.
- μ_0 is outside the CI iff $|\bar{X} - \mu_0| > z_{\alpha/2} \cdot \text{SE}$ (same condition).

9.1 Two Sides of the Same Coin 2

Picture. Plot the CI as a bar on the number line and mark μ_0 :



Practical reading. Once you have a 95% CI, you immediately know which two-sided hypothesis tests would reject at $\alpha = 0.05$: every μ_0 outside the CI rejects; every μ_0 inside the CI fails to reject.

Reverse direction. Conversely, every two-sided test gives you a CI “for free”: collect all μ_0 values that the test fails to reject — that’s the $(1 - \alpha) \cdot 100\%$ CI.

10. Summary

10.1 Take Away — Foundations

From point to set estimation

- A **point estimate** ($\bar{X}$) is a single number; it tells you nothing about uncertainty.
- A **confidence interval** reports a *range* of plausible values for μ together with a confidence level $1 - \alpha$.
- Interpretation: the *procedure* catches μ in $(1 - \alpha) \cdot 100\%$ of repeated samples. The $1 - \alpha$ is about the method, not any single interval.
- Large-sample approach: for $n \geq 60$, treat $(\bar{X} - \mu)/(s/\sqrt{n})$ as approximately $N(0, 1)$. No t needed.

10.2 Take Away — CI Recipe (Exam Target)

Confidence interval for μ

$$\text{CI} = \bar{X} \pm z_{\alpha/2} \cdot \frac{s}{\sqrt{n}} \quad (\text{or } \sigma/\sqrt{n} \text{ if } \sigma \text{ known}).$$

Critical values to memorize:

$$z_{0.05} = 1.645, \quad z_{0.025} = 1.96, \quad z_{0.005} = 2.576.$$

Five-step recipe: (1) compute $\bar{X}$, s ; (2) pick α , look up $z_{\alpha/2}$; (3) SE; (4) $\text{ME} = z_{\alpha/2} \cdot \text{SE}$; (5) report $[\bar{X} \pm \text{ME}]$ with interpretation.

10.3 Take Away — HT Recipe (Exam Target)

Hypothesis test for $\mu = \mu_0$

Test statistic: $Z = (\bar{X} - \mu_0)/(s/\sqrt{n})$, approx. $N(0, 1)$ under H_0 for $n \geq 60$.

- **Critical-value method:** reject if $|Z| > z_{\alpha/2}$ (two-sided), $Z > z_{\alpha}$ (right-tailed), or $Z < -z_{\alpha}$ (left-tailed).
- **P-value method:** reject if $p \leq \alpha$, where $p = 2\mathbb{P}(Z > |z_{\text{obs}}|)$ for two-sided.

Both methods give the same decision.

Significance level $\alpha = \mathbb{P}(\text{Type I error}) = \text{false-alarm rate}$. Default $\alpha = 0.05$.

10.4 Where Each Tool Reappears

Forward pointers

- **CI / HT recipes** — applied to OLS coefficients (Lecture 7):
 $\hat{\beta} \pm 1.96 \cdot \widehat{SE}(\hat{\beta})$ for a 95% CI; reject $H_0 : \beta = 0$ if $|\hat{\beta}/\widehat{SE}(\hat{\beta})| > 1.96$.
- **CI $\leftrightarrow$ HT equivalence** — read significance off a coefficient table by checking whether 0 is inside the CI (Lecture 7).
- **Large-sample Normal approach** — regression inference in Lecture 7 uses Z, not t. Same critical values 1.96, 2.576.
- **p-values** — the standard summary of evidence in every regression table, every classification accuracy test, and every A/B testing report in industry.

Next time: prediction and inference — the overarching framework that motivates the rest of the course.